

# Matrix Computation Homework 4

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## Problem 1. (Quadratic Form)

1)

**Solution.**

It is obviously that:

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & a \end{bmatrix}$$

2)

**Solution.**

First, we find the eigenvalues by solving the characteristic equation  $|\mathbf{A} - \lambda \mathbf{I}| = 0$ :

$$\det \begin{bmatrix} 2 - \lambda & 1 & 1 \\ 1 & 2 - \lambda & 1 \\ 1 & 1 & a - \lambda \end{bmatrix} = 0$$

It is obviously that:

$$(1 - \lambda)(\lambda^2 - (3 + a)\lambda + 3a - 2) = 0$$

Thus we can get that  $\lambda_1 = 1$  and  $\lambda_2 + \lambda_3 = 3a - 2$ ,  $\lambda_2 \cdot \lambda_3 = (3 + a)/2$

Since  $\mathbf{A}$  is PSD, we need to ensure:

$$\lambda_2 + \lambda_3 \geq 0 \quad \text{and} \quad \lambda_2 \cdot \lambda_3 \geq 0$$

Thus we can get:

$$a \geq \frac{2}{3}$$

3)

**Solution.**

Since  $a = 2$ ,

$$(1 - \lambda)(\lambda^2 - (3 + a)\lambda + 3a - 2) = 0 \implies \lambda_1 = \lambda_2 = 1, \lambda_3 = 4$$

For  $\lambda = 1$ , we can easily get

$$\mathbf{w}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{w}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Apply Gram-schmidt to  $\mathbf{w}_1, \mathbf{w}_2$ :

$$\begin{aligned} \mathbf{u}_1 &= \mathbf{w}_1 \\ \mathbf{u}_2 &= \mathbf{w}_2 - \frac{\mathbf{w}_2^T \mathbf{u}_1}{\mathbf{u}_1^T \mathbf{u}_1} \mathbf{u}_1 = \begin{bmatrix} 0.5 \\ 0.5 \\ -1 \end{bmatrix} \end{aligned}$$

Normalize the vector and get:

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{q}_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

For  $\lambda = 4$ , we can easily get

$$\mathbf{q}_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Thus,

$$\begin{aligned} \mathbf{A}^{1/2} &= \sqrt{\lambda_1} \mathbf{q}_1 \mathbf{q}_1^T + \sqrt{\lambda_2} \mathbf{q}_2 \mathbf{q}_2^T + \sqrt{\lambda_3} \mathbf{q}_3 \mathbf{q}_3^T \\ &= 1 \cdot \mathbf{q}_1 \mathbf{q}_1^T + 1 \cdot \mathbf{q}_2 \mathbf{q}_2^T + 2 \cdot \mathbf{q}_3 \mathbf{q}_3^T \\ &= \begin{bmatrix} 4/3 & 1/3 & 1/3 \\ 1/3 & 4/3 & 1/3 \\ 1/3 & 1/3 & 4/3 \end{bmatrix} \end{aligned}$$

## Problem 2. (Positive Semidefinite Matrices)

1)

**Solution.**

**Proof symmetric:**

$$A^T = (I + \alpha\alpha^T)^T = I^T + (\alpha\alpha^T)^T = I + (\alpha^T)^T\alpha^T = I + \alpha\alpha^T = A$$

**Proof PD:**

$$\begin{aligned}\mathbf{x}^T \mathbf{A} \mathbf{x} &= \mathbf{x}^T (I + \alpha\alpha^T) \mathbf{x} \\ &= \mathbf{x}^T \mathbf{x} + (\mathbf{x}^T \alpha)(\alpha^T \mathbf{x}) \\ &= \|\mathbf{x}\|_2^2 + \|\alpha^T \mathbf{x}\|_2^2\end{aligned}$$

Thus, when  $\mathbf{x} \neq 0$ ,  $\mathbf{x}^T \mathbf{A} \mathbf{x} > 0$ , i.e.  $\mathbf{A}$  is PD matrix.

**2)**

**Solution.**

**Case 1: The eigenvector is parallel to  $\alpha$**

$$\begin{aligned}A\alpha &= (I + \alpha\alpha^T)\alpha \\ &= \alpha + \alpha(\alpha^T\alpha) \\ &= (1 + \beta)\alpha\end{aligned}$$

Thus,  $\lambda_1 = 1 + \beta$  is an eigenvalue with eigenvector  $\alpha$ . Since  $\alpha$  is a single vector, the algebraic multiplicity is 1.

**Case 2: The eigenvectors are orthogonal to  $\alpha$**

Let  $v$  be any non-zero vector satisfy  $\alpha^T v = 0$ .

$$\begin{aligned}Av &= (I + \alpha\alpha^T)v \\ &= v + \alpha(\alpha^T v) \\ &= v\end{aligned}$$

Thus,  $\lambda_2 = 1$  is an eigenvalue.

Since the subspace orthogonal to  $\alpha$  has dimension  $n - 1$ , there are  $n - 1$  linearly independent eigenvectors associated with this eigenvalue. The algebraic multiplicity is  $n - 1$ .

**3)**

**Solution.**

It is obviously that:

$$\begin{aligned} C &= (\alpha\alpha^T)^3 \\ &= \beta^2\alpha\alpha^T \\ &= (\beta\alpha)(\beta\alpha)^T \end{aligned}$$

Thus,

$$\begin{aligned} C^{1/2} &= \frac{1}{\|\beta\alpha\|_2}(\beta\alpha)(\beta\alpha)^T \\ &= \frac{\beta^2}{\sqrt{(\beta\alpha)^T(\beta\alpha)}}\alpha\alpha^T \\ &= \sqrt{\beta}\alpha\alpha^T \end{aligned}$$

### Problem 3. (Application of Schur Complement)

**Solution.**

We will use the Schur Complement to solve this problem.

$$M = \begin{bmatrix} A & B \\ B^T & C \end{bmatrix}$$

Then if  $A \succ 0$ , we just need

$$S = C - B^T A^{-1} B \succeq 0$$

and we will get  $M \succeq 0$ .

To reduce the computation, we move rows and columns containing  $a$  to the end.

Then we can get

$$X = \begin{bmatrix} 2 & \mathbf{2} & 1 & 0 & 0 & -1 \\ \mathbf{2} & \mathbf{a} & \mathbf{2} & \mathbf{0} & \mathbf{0} & \mathbf{2} \\ 1 & \mathbf{2} & 2 & 0 & 0 & 0 \\ 0 & \mathbf{0} & 0 & 2 & 0 & 0 \\ 0 & \mathbf{0} & 0 & 0 & 2 & 0 \\ -1 & \mathbf{2} & 0 & 0 & 0 & 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ -1 & 0 & 0 & 0 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 2 & 0 & 0 & 2 \end{bmatrix}^T$$

$$C = a$$

To make  $X$  be PSD matrix, we just need to make sure  $A \succ 0$  and  $S = C - B^T A^{-1} B \succeq 0$ .

**Step1. Prove  $A \succ 0$**

Since the 3rd, 4th row of  $A$  only have diagonal entry 2, we ignore them to simplify the problem.

Then we focus on

$$A_{core} = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 0 \\ -1 & 0 & 2 \end{bmatrix}$$

Since:

- $2 > 0$
- $\det \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = 3 > 0$
- $\det \begin{pmatrix} 2 & 1 & -1 \\ 1 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix} = 4 > 0$

we can get that  $A \succ 0$ .

**Step2. Prove  $S = C - B^T A^{-1} B \succeq 0$**

Let  $y = A^{-1} B$ , then  $Ay = B$ .

We can solve the line system and get that:

$$y = [2, 0, 0, 0, 2]^T$$

Thus,

$$\begin{aligned} S &= C - B^T A^{-1} B \\ S &= a - B^T y \\ &= a - \begin{bmatrix} 2 & 2 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \\ 2 \end{bmatrix} \\ &= a - 8 \end{aligned}$$

If we want  $S \geq 0$ , we just need  $a \geq 8$

## Problem 4. (Matrix Inequality, Matrix Norm)

1)

**Solution.**

First, we can easily get

$$\sigma_{\min}(\mathbf{X}) = \min_{\|\mathbf{v}\|_2=1} \|\mathbf{X}\mathbf{v}\|_2$$

Since  $\mathbf{X}^\top \mathbf{X} - \mathbf{I}_r$  is a real symmetric matrix, we can get:

$$\|\mathbf{X}^\top \mathbf{X} - \mathbf{I}_r\|_2 = \max_{\|\mathbf{v}\|_2=1} |\mathbf{v}^\top (\mathbf{X}^\top \mathbf{X} - \mathbf{I}_r) \mathbf{v}| = \tau$$

Then,

$$\begin{aligned} \max_{\|\mathbf{v}\|_2=1} |\mathbf{v}^\top (\mathbf{X}^\top \mathbf{X} - \mathbf{I}_r) \mathbf{v}| &= \tau \\ |\mathbf{v}^\top (\mathbf{X}^\top \mathbf{X} - \mathbf{I}_r) \mathbf{v}| &\leq \tau \\ |\mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} - 1| &\leq \tau \\ \|\mathbf{X}\mathbf{v}\|_2^2 - 1 &\leq \tau \end{aligned}$$

Thus,

$$1 - \tau \leq \|\mathbf{X}\mathbf{v}\|_2^2 \leq 1 + \tau$$

Since  $1 - \tau \leq \|\mathbf{X}\mathbf{v}\|_2^2$ , we can get:

$$\min_{\|\mathbf{v}\|_2=1} \|\mathbf{X}\mathbf{v}\|_2^2 \geq 1 - \tau$$

i.e.,

$$\sigma_{\min}(\mathbf{X})^2 \geq 1 - \tau$$

$$\sigma_{\min}(\mathbf{X}) \geq \sqrt{1 - \tau}$$

Since  $0 \leq \tau < 1$ , we know that  $\sqrt{1 - \tau} \geq 1 - \tau$ .

Therefore:

$$\sigma_{\min}(\mathbf{X}) \geq 1 - \tau$$

2)

**Solution.**

$$\sigma_1(\mathbf{E}) = \sqrt{\lambda_{\max}(\mathbf{E}^\top \mathbf{E})}$$

First, we get  $\mathbf{E}^\top \mathbf{E}$

$$\begin{aligned}\mathbf{E}^\top \mathbf{E} &= (\mathbf{a}\mathbf{b}^\top)^\top (\mathbf{a}\mathbf{b}^\top) \\ &= \|\mathbf{a}\|_2 \mathbf{b}\mathbf{b}^\top\end{aligned}$$

Since  $\mathbf{E}^\top \mathbf{E} = \|\mathbf{a}\|_2 \mathbf{b}\mathbf{b}^\top$  is a rank-1 matrix, we can get that  $\mathbf{E}^\top \mathbf{E}$  only have one none zero eigenvalue.

Consider the vector  $\mathbf{b}$  as a candidate eigenvector for the matrix  $\mathbf{b}\mathbf{b}^\top$ :

$$(\mathbf{b}\mathbf{b}^\top)\mathbf{b} = \mathbf{b}(\mathbf{b}^\top \mathbf{b}) = \mathbf{b}(\|\mathbf{b}\|_2^2) = \|\mathbf{b}\|_2^2 \mathbf{b}$$

This shows that  $\mathbf{b}$  is an eigenvector of  $\mathbf{b}\mathbf{b}^\top$  with respect to the eigenvalue  $\|\mathbf{b}\|_2^2$ .

Thus, the non-zero eigenvalue of the full matrix  $\mathbf{E}^\top \mathbf{E} = \|\mathbf{a}\|_2^2 (\mathbf{b}\mathbf{b}^\top)$  is:

$$\lambda_{\max} = \|\mathbf{a}\|_2^2 \cdot \|\mathbf{b}\|_2^2$$

$$\sigma_1(\mathbf{E}) = \sqrt{\lambda_{\max}(\mathbf{E}^\top \mathbf{E})} = \|\mathbf{a}\|_2 \cdot \|\mathbf{b}\|_2$$

## Problem 5. (SVD)

**Solution.**

First, we get  $AA^\top$ :

$$AA^\top = \begin{bmatrix} \cos(\theta_u) & \sin(\theta_u) \\ \cos(\theta_v) & \sin(\theta_v) \end{bmatrix} \begin{bmatrix} \cos(\theta_u) & \cos(\theta_v) \\ \sin(\theta_u) & \sin(\theta_v) \end{bmatrix} = \begin{bmatrix} 1 & \cos(\delta) \\ \cos(\delta) & 1 \end{bmatrix}$$

which  $\delta = \theta_v - \theta_u$

Next, find the eigenvalues  $\lambda$  of  $AA^\top$ .

We solve  $\det(\lambda I - AA^\top) = 0$  and get

$$(\lambda - 1)^2 - \cos^2(\delta) = 0 \implies \lambda - 1 = \pm \cos(\delta) \implies \lambda = 1 \pm \cos(\delta)$$

Thus,

$$\begin{aligned}\sigma_1 &= \sqrt{1 + \cos(\delta)} = \sqrt{2 \cos^2\left(\frac{\delta}{2}\right)} = \sqrt{2} \cos\left(\frac{\theta_v - \theta_u}{2}\right) \\ \sigma_2 &= \sqrt{1 - \cos(\delta)} = \sqrt{2 \sin^2\left(\frac{\delta}{2}\right)} = \sqrt{2} \sin\left(\frac{\theta_v - \theta_u}{2}\right)\end{aligned}$$

and the matrix  $\Sigma$  is:

$$\Sigma = \begin{bmatrix} \sqrt{2} \cos\left(\frac{\theta_v - \theta_u}{2}\right) & 0 \\ 0 & \sqrt{2} \sin\left(\frac{\theta_v - \theta_u}{2}\right) \end{bmatrix}$$

For  $\lambda_1 = 1 + \cos(\delta)$ :

$$\begin{bmatrix} 1 - (1 + \cos \delta) & \cos \delta \\ \cos \delta & 1 - (1 + \cos \delta) \end{bmatrix} \mathbf{u}_1 = \mathbf{0} \implies \begin{bmatrix} -\cos \delta & \cos \delta \\ \cos \delta & -\cos \delta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{0}$$

It is obviously that

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For  $\lambda_2 = 1 - \cos(\delta)$ :

$$\begin{bmatrix} \cos \delta & \cos \delta \\ \cos \delta & \cos \delta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{0}$$

It is obviously that

$$\mathbf{u}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Thus, the matrix  $U$  is:

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Finally, we find  $V$ :

We use the relation  $AV = U\Sigma$ , which implies  $v_i = \frac{1}{\sigma_i} A^T u_i$ .

$$A^T u_1 = \begin{bmatrix} \cos \theta_u & \cos \theta_v \\ \sin \theta_u & \sin \theta_v \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} \cos \theta_u + \cos \theta_v \\ \sin \theta_u + \sin \theta_v \end{bmatrix} = \sqrt{2} \cos\left(\frac{\theta_v - \theta_u}{2}\right) \begin{bmatrix} \cos\left(\frac{\theta_u + \theta_v}{2}\right) \\ \sin\left(\frac{\theta_u + \theta_v}{2}\right) \end{bmatrix}$$

Thus,

$$\mathbf{v}_1 = \frac{1}{\sigma_1} A^T u_1 = \begin{bmatrix} \cos\left(\frac{\theta_u + \theta_v}{2}\right) \\ \sin\left(\frac{\theta_u + \theta_v}{2}\right) \end{bmatrix}$$

Similarly, we can get:

$$A^T u_2 = \begin{bmatrix} \cos \theta_u & \cos \theta_v \\ \sin \theta_u & \sin \theta_v \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} \cos \theta_u - \cos \theta_v \\ \sin \theta_u - \sin \theta_v \end{bmatrix} = \sqrt{2} \sin\left(\frac{\theta_v - \theta_u}{2}\right) \begin{bmatrix} \sin\left(\frac{\theta_u + \theta_v}{2}\right) \\ -\cos\left(\frac{\theta_u + \theta_v}{2}\right) \end{bmatrix}$$

and

$$\mathbf{v}_2 = \begin{bmatrix} \sin\left(\frac{\theta_u + \theta_v}{2}\right) \\ -\cos\left(\frac{\theta_u + \theta_v}{2}\right) \end{bmatrix}$$

Finally, the SVD  $A = U\Sigma V^T$  is:

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \sqrt{2} \cos\left(\frac{\theta_v - \theta_u}{2}\right) & 0 \\ 0 & \sqrt{2} \sin\left(\frac{\theta_v - \theta_u}{2}\right) \end{bmatrix}$$

$$V^T = \begin{bmatrix} \cos\left(\frac{\theta_u + \theta_v}{2}\right) & \sin\left(\frac{\theta_u + \theta_v}{2}\right) \\ \sin\left(\frac{\theta_u + \theta_v}{2}\right) & -\cos\left(\frac{\theta_u + \theta_v}{2}\right) \end{bmatrix}$$



## Problem 6. (Sensitivity in Linear Systems)

1)

**Solution.**

Since  $\Delta x = A^\dagger \Delta y$ , we can get that  $\|\Delta x\|_2 \leq \|A^\dagger\|_2 \|\Delta y\|_2$ .

Thus,

$$\frac{\|\Delta x\|_2}{\|x\|_2} \leq \frac{\|A^\dagger\|_2 \|\Delta y\|_2}{\|x\|_2}$$

Then apply the hint and get:

$$\frac{\|\Delta x\|_2}{\|x\|_2} \leq \frac{\|A^\dagger\|_2 \|\Delta y\|_2 \|A\|_2}{\|y\|_2} = \kappa_2(A) \frac{\|\Delta y\|_2}{\|y\|_2}$$

2)

**Solution.**

Use the hint, we can get:

$$\|\Delta x\|_2 = \|A^\dagger \Delta y\|_2 \leq \|A^\dagger\|_2 \|\Delta y\|_2$$

Then use the inequality  $\|A^\dagger\|_2 \leq \|A_1^{-1}\|_2$ , we can get:

$$\|\Delta x\|_2 \leq \|A_1^{-1}\|_2 \|\Delta y\|_2$$

Since  $\|A_1^{-1}\|_2$  and  $\|\Delta y\|_2$  are not relate to  $m$ , we can get the upper bound of  $\|\Delta x\|_2$  cannot infinitely increase with the increase of  $m$ .